

Lecture 8 Discriminative Learning - Linear Classifier

Generative models:

1. Learn class conditional density (CCD) from the data: $p(x|y)$
2. Use Bayesian decision rule (BDR) to get the classifier: $p(y|x)$

The data is used to learn the CCD

The classifier is a byproduct (secondary goal)

Density estimation is hard

Data $\xrightarrow{\text{ill-posed}}$ CCD $\rightarrow$ Classifier

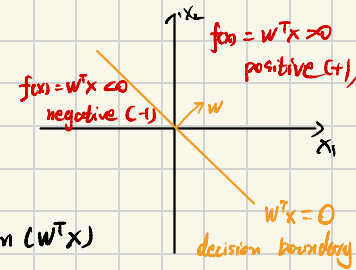
Discriminative learning is solve the decision boundary, the classifier from the data directly

Linear classifier

binary classes: $\{-1, +1\}$ $x \in \mathbb{R}^d$

function $f(x) = w^T x$ $w \in \mathbb{R}^d$

Decision Rule: $y = \begin{cases} +1 & w^T x > 0 \\ -1 & w^T x < 0 \end{cases} = \text{sign}(w^T x)$

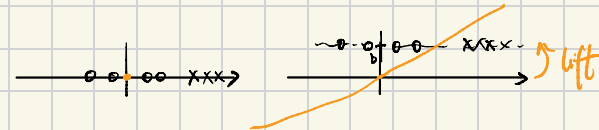


bias

$$\tilde{x} = \begin{bmatrix} x \\ 1 \end{bmatrix}$$

$$f(\tilde{x}) = \tilde{w}^T \tilde{x} = w^T x + b$$

$$\tilde{w} = \begin{bmatrix} w \\ b \end{bmatrix}$$



Lecture 8 Discriminative Learning - Linear Classifier

Training Set $D = \{(x_i, y_i)\}_{i=1}^n$ $D = \{X, y\}$ $X = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$ $y = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}$

Note:

given w

$$\underbrace{y_i \cdot w^T x_i}_{\substack{>0 \\ <0}} > 0 \Rightarrow x_i \text{ correct classified.}$$

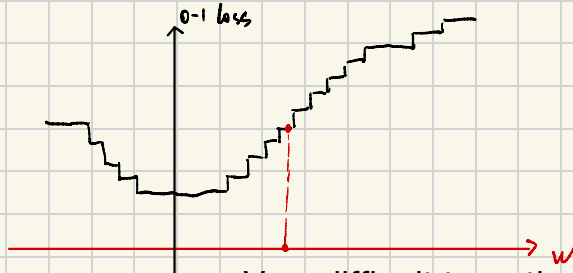
$$\underbrace{y_i \cdot w^T x_i}_{\substack{>0 \\ <0}} < 0 \Rightarrow x_i \text{ mis classified.}$$

Ideal case 0-1 loss func

$$w^* = \arg \min_w \sum_{i=1}^n \begin{cases} 0, & y_i \cdot w^T x_i > 0 \\ 1, & y_i \cdot w^T x_i < 0 \end{cases}$$

correct
incorrect

Minimize the number of misclassifications



Very difficult to optimize
Hard to define the gradient

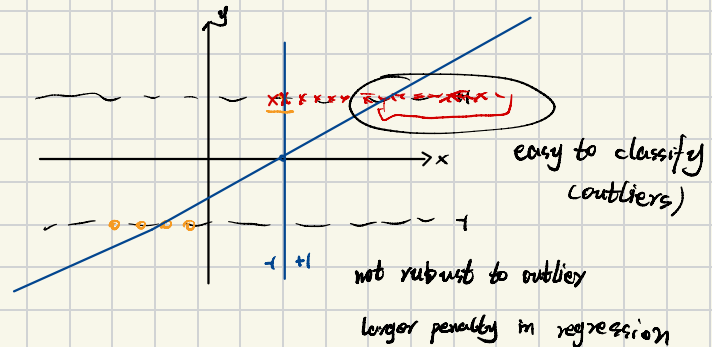
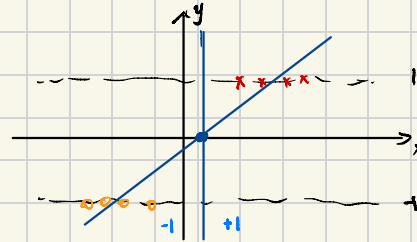
Least Square Classification (Label regression)

Ignore the fact that y is discrete, just model it as a least square regression

Training set $D = \{X, y\}$

$$\text{LSC } w^* = \arg \min_w \sum_i (w^T x_i - y_i)^2 = \arg \min_w \|X^T w - y\|^2$$

$$w^* = (X X^T)^{-1} X y$$



Made with Goodnotes

Perceptron criteria: only look at the misclassified points

$$E(w) = \sum_{i \in M} y_i w^T x_i$$

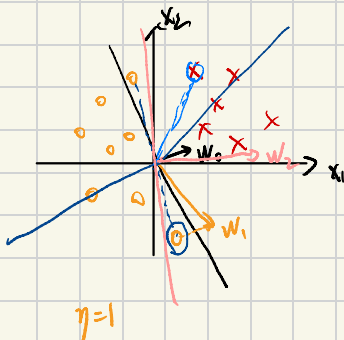
misclassified.

$$\{y_i w^T x_i < 0\}$$

Target $Z_{CW} = 0 \rightarrow$ Training set correctly classified

Goal: $w^* = \arg\min_w Z(w) = \arg\min_w \sum_{i \in M} -y_i w^T x_i$

$$w_{t+1} = w_t + \eta y_i x_i$$



update w

- 1) rotate towards positive class

- 2) length of w $\nearrow$

effect of each point
in updating $\downarrow$

converge within $\left(\frac{R}{r}\right)^2$ iterations

$$R := \max_i \|x_i\|$$

γ : margin if $\|w\| = 1$, $y_i w^T x_i \geq \gamma \quad \forall i$

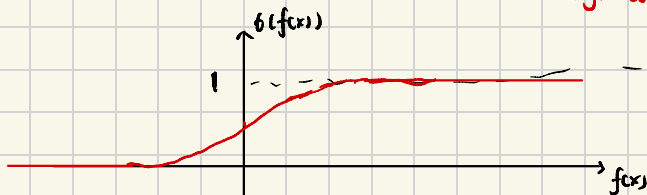


Lecture 8 Discriminative Learning - Linear Regression (Probabilistic Approach)

$$y \in \{0, 1\}$$

consider BPR, if CCD is Gaussian. $p(x|y) = \mathcal{N}(x|y, \Sigma_y)$

posterior: $p(y=1|x) = \frac{1}{1 + e^{-f(x)}} = b(f(x))$ sigmoid func



$$\begin{aligned} f(x) \gg 0 & \quad b(f(x)) = 1 \\ f(x) \ll 0 & \quad b(f(x)) = 0 \end{aligned}$$

$$f(x) = w^T x$$

$$p(y=1|x) = b(w^T x) = \frac{1}{1 + e^{-w^T x}} = \pi$$

Decision $y = \begin{cases} 1 & p(y=1|x) > \frac{1}{2} \\ 0 & \text{otherwise} \end{cases} = \begin{cases} 1 & w^T x > 0 \\ 0 & w^T x < 0 \end{cases}$

Learning

$$D = \{(x_i, y_i)\}_{i=1}^n$$

$$\pi_i = b(w^T x_i) = p(y_i=1|x_i, w)$$

$$p(y_i|x_i, w) = \pi_i^{y_i} (1 - \pi_i)^{1-y_i}$$

↑ ↑ ↑
class input wgt

likelihood (Bernoulli)

log likelihood

$$\mathcal{L}(w) = \sum_{i=1}^n \log p(y_i|x_i, w) = \sum_{i=1}^n y_i \log \pi_i + (1 - y_i) \log (1 - \pi_i)$$

MLZ

$$w^* = \arg \max_w \mathcal{L}(w)$$

Alternatively

$$w^* = \arg \min_w -\mathcal{L}(w) = \arg \min_w \sum_{i=1}^n -y_i \log \pi_i - (1 - y_i) \log (1 - \pi_i)$$

binary Cross Entropy Loss

maximize $J(w)$ via gradient ascent

find zero crossing of gradient

$$w_{t+1} = w_t - \underbrace{(\nabla^2 J(w))}^{\text{Hessian}} \underbrace{(\nabla J(w))}^{\text{grad}}$$

$$w_{t+1} = (X^T R X)^{-1} X^T R z \quad \leftarrow \begin{array}{l} \text{weighted least squares} \\ R, z, X \end{array}$$

where

$$R = \text{diag}(\pi_1(1-\pi_1) \dots \pi_n(1-\pi_n)) \quad \leftarrow \text{weight most depends on } w$$

$$z = \underbrace{X^T w_t}_{f_t(x)} - \underbrace{R^{-1}(\pi - y)}_{\text{pred-ground truth}}$$

iterative reweighted least square (IRWLS / IRLS)

Comparison of the Loss functions

All methods are of the form

$$w^* = \arg \min_w \sum_{i=1}^n \underbrace{L(f(x_i), y_i)}_{\text{ERM}} \quad \text{empirical risk}$$

Let $z_i = y_i w^T x_i$

1) 0-1 loss : $L = \begin{cases} 0 & z_i > 0 \\ 1 & z_i < 0 \end{cases}$

2) LSL : $L = (z_i - 1)^2$

3) perceptron : $L = \begin{cases} 0 & z_i > 0 \\ -z_i & z_i < 0 \end{cases} = \max(0, -z_i)$

4) LR : $L = \left[\log(1 + e^{-z_i}) \right] \cdot \frac{1}{\log 2}$

